

STRAY SCANNER -> COZMO-AI-V2

ONE-PAGE CAPTURE PROTOCOL

Print it. Follow every step. [OFFICIAL] = app fact; [PROJECT] = capture rule.

Checked 2026-09-02

Capture log: Site/location _____ Floor/area _____ Operator _____ Date/time _____ Marker _____ m

STOP RULE

If an avoid/restart condition occurs, stop, mark the folder BAD, fix the room, and start a new capture at the marker. Never resume a failed file or join captures: one pipeline run is one continuous coordinate frame.

1 | INSTALL + DEVICE

[OFFICIAL] Install **Stray Scanner** (Kenneth Blomqvist, free) from the App Store; grant Camera. Current listing: iOS 18.6+.

[OFFICIAL] Use LiDAR: iPhone 12 Pro/Pro Max or later Pro, or iPad Pro 11-inch 2nd gen+ / 12.9-inch 4th gen+. No non-Pro substitute. If iPad install is refused, use an eligible iPhone.

[PROJECT] Leave 10 GB free; charge to 80% or connect power; close apps; Do Not Disturb; Auto-Lock Never; silence calls/notifications.

2 | PREPARE THE ROOM

[PROJECT] Turn on even lighting; avoid direct sun, flicker, glare, and dark corners. Open curtains/blinds so jambs show. Cover or flag mirrors/reflective glass; never rely on them as the only wall evidence.

Open and wedge every door. Remove people/pets. Clear small clutter from wall/floor views; do not move furniture, doors, or curtains after Record.

Tape a straight, non-reflective measured marker (prefer 2.00 m); write its exact length. Keep it fixed and show both endpoints at start and finish. It validates scale later; it does not auto-calibrate.

3 | APP SETTINGS + PREFLIGHT

Open **Record new session**. Tap the fps control until it reads **30 fps**. **[PROJECT]** 30 fps is recommended; the app also offers 60/15/5/1. Keep portrait. Make a 5-second test: Record, hold still, Stop, open the test, Share -> Save to Files. Confirm it opens and contains rgb.mp4, depth/, confidence/, odometry.csv, imu.csv, camera_matrix.csv. Delete test only after pass.

[OFFICIAL] Contract: depth PNG = 16-bit millimetres; confidence PNG = 0/1/2; matching six-digit stems index RGB frames; odometry/IMU timestamps seconds, odometry positions metres, IMU accel m/s² and angular rate rad/s, camera_matrix 3x3 pixel intrinsics.

[PROJECT] Nonblank odometry timestamp rows = depth PNG count = confidence PNG count; decoded RGB count should match. Any mismatch, unreadable PNG, or terminal video frame loss = FAIL and recapture.

4 | HOLD THE DEVICE

[PROJECT] Portrait; screen toward you; rear camera/LiDAR toward scene; top edge up; no roll. Keep the lens about **1.4 m above floor** (within about 10 cm) for the run.

Keep level for wall sweeps. Only at floor/ceiling pauses pitch about 15 deg down/up, then return to level. Never turn sideways/upside down.

Keep surfaces about **1-3 m** away where possible; never intentionally rely on > **3.5 m**. Walk closer in large rooms. Keep both windows uncovered.

5 | EXACT ROUTE

[PROJECT] Use this route exactly.

1. Start at marker. Tap Record. Hold full marker motionless **3 s**.
2. Walk clockwise, wall on left. Advance no more than **0.5 m every 2 s**; pause **2 s at every corner**. Keep knee-to-head wall band visible.

3. Doorway: approach level; hold **2 s** with both jambs, header, threshold visible; cross without height/orientation change; hold **2 s just inside**; sweep room clockwise.

4. Each room: every wall/corner; centre holds **2 s level + 2 s down for floor strip + 2 s up for ceiling edge**; each window **2 s face-on** with jambs, sill, header.

5. Return through doors to within **2 m** of marker after at least **4 s**. Hold marker **3 s**, then Stop. **[PROJECT]** This revisit enables loop closure.

6 | DURATION + SPLIT

[PROJECT] Target **30-120 s** per connected floor/area. Hard operator cap: **5 min** per continuous capture. This is a project safety rule, not an App Store limit; it limits drift, interruption, and terminal loss.

If it will exceed 5 min, cross floors, or enter a disconnected area, stop at a marker and start a new capture with its own marker/folder. **Never join folders.**

7 | AVOID / RESTART

[PROJECT] Stop and restart from marker if: portrait/roll changes; whip turn or blur; lens/LiDAR covered; call, lock, app interruption, or Record stops; AR view jumps/loses tracking; room missed; surface >3.5 m; door/furniture/curtain moves; people/pets enter; or mirror/glass is only evidence.

Capture solid adjacent surfaces and both jambs; flag unavoidable reflective spans for manual measurement. Do not resume a failed file.

8 | STOP + EXPORT + HANDOFF

[OFFICIAL] After Stop, recording detail -> **Share** -> Save to Files or AirDrop; keep app ZIP unchanged. Original-folder route: cable -> Finder -> device -> **Files** -> **Stray Scanner** -> drag dataset folder (Windows: iTunes). Files route: Browse -> On My iPhone/iPad -> Stray Scanner -> Share/Save.

[PROJECT] Name only the delivered top folder site_floor_area_YYYYMMDD-HHMM; retain app hash in note. Do not rename children, rotate/transcode video, resize PNGs, reorder files, or rearrange folders. distortion/ is optional; all other listed items are required.

Before transfer, recheck required names and equal counts. ZIP: unzip -t capture.zip, then shasum -a 256 capture.zip; send original ZIP + checksum. Do not unzip/repack unless asked.

9 | CURRENT-MAIN PROCESS

From repo root, after uv sync (verified 2026-09-02):

```
uv run python -m cozmo_ai_v2.pipeline run "/path/to/CAPTURE" \
--out "out/CAPTURE" --no-damage
Expect result.json, floorplan.svg, scene.glb, cloud.ply, mesh.ply,
planes.json, fusion_manifest.json, and scope/openings CSVs.
Recapture for terminal sidecar unavailable, requested frames rejected, low IMU
gravity consistency, floor not observed/low confidence, no ceiling, no rooms, or
room-overlap warnings. Uncalibrated intervals and reference-not-applied are
review/scale notes. Defaults: confidence >=1, depth <=3.5 m; invalid depth is
discarded.
```

10 | FINAL 20-SECOND PASS/FAIL

[] LiDAR device; 10 GB free; 80%/power; no interruptions

[] 30 fps; portrait/no roll; lens 1.4 m high; 1-3 m surfaces

[] Measured marker at start/finish; all rooms, doors, corners, windows

[] Floor strip + ceiling edge; no people/pets/moving objects; revisit

[] Original folder/ZIP; names/counts match; checksum recorded

[] Pipeline output has no recapture warning